



Food and Agriculture Organization
of the United Nations

FAO Ethiopia



Leveraging Evidence for Effective Anticipatory Action Workshop
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Mitigating the impacts of anticipated drought on vulnerable pastoral communities in Ethiopia

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Presentation outline

- Introduction
- Intervention packages
- Theory of change
- Assessment objectives
- Methods
- Results
- Conclusions
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Introduction

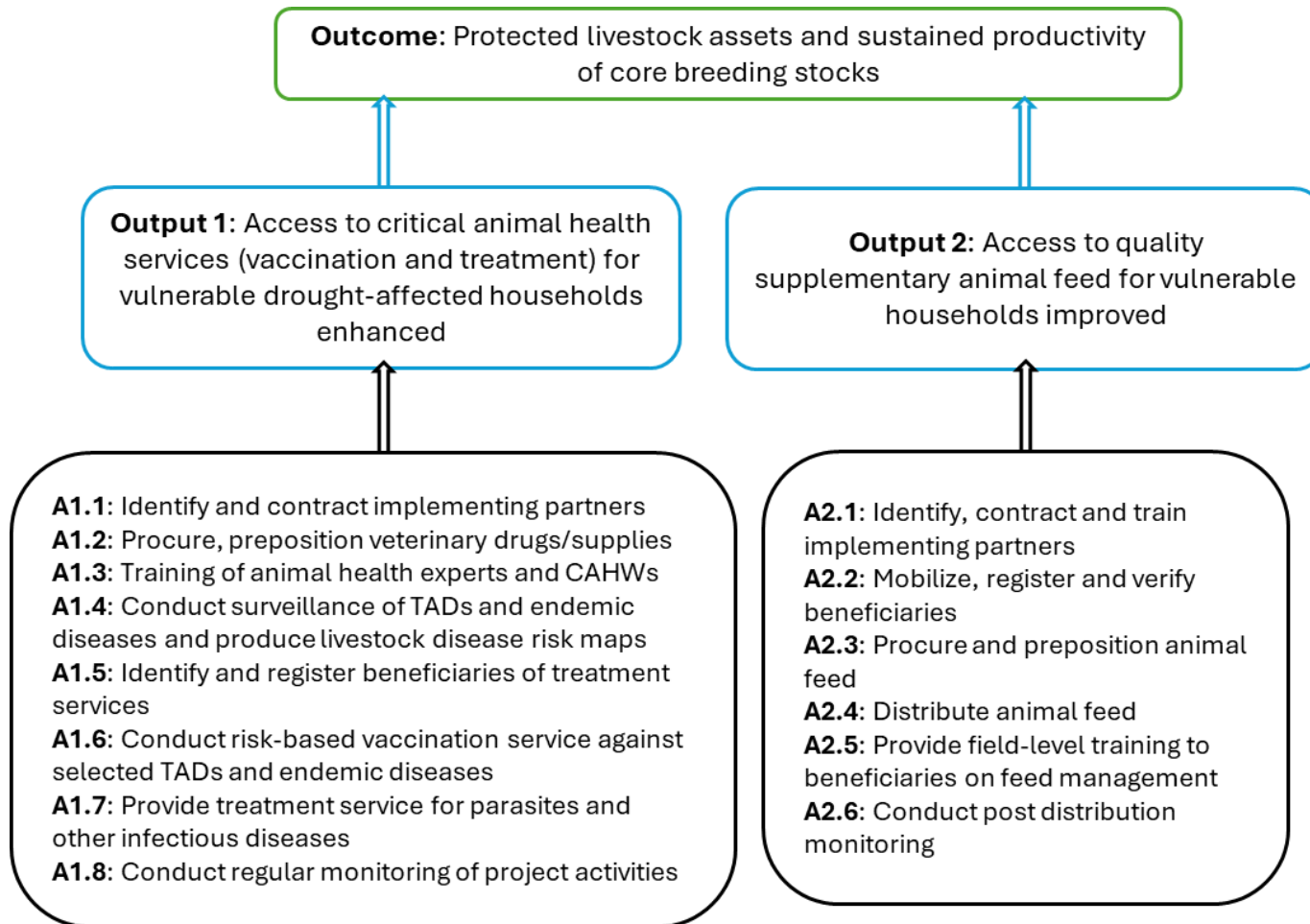
- Pastoral lowland of Oromia and Somali suffered climatic shocks, the 2020–2022 droughts killed ~ 7 million livestock (costed ~USD 1.5 billion)
- Favorable rainfall in 2024 led to improvements in agricultural outputs
- ICPAC forecast and EMI early warnings predicted drought risk in Oromia & Somali regions due to delayed and poor Oct–Dec 2024 rainfall – negatively affect MAM season (key for rangeland regeneration, crop production & livelihoods)
 - ✓ Anticipatory Action (AA) triggered at the end of Feb 2025
- FAO implemented AA to mitigate impacts of drought on pastoral livelihoods within the AA window for livestock-based livelihoods (b/n March–August)

Intervention packages

- Animal health services:
 - ✓ **Vaccinated** animals: 6,169,714 (792,395 cattle & 5,377,319 SRs); **66,524 HHs** (~56% female-headed)
 - ✓ **Treated** animals: 195,000 (45,000 cattle & 150,000 SRs); **15,300 HHs** (~30% female-headed)
- Supplementary animal feed (TMR):
 - ✓ 4000 HHs in Somali (63.6% female-headed)
- Capacity-building training on animal health:
 - ✓ 140 animal health staff (44% female) and 210 CAHWs (32% female)
- Early warning information



Theory of change



Assessment objectives

- To evaluate the contribution of the AA interventions in:
 - ✓ Protecting livestock assets and productivity
 - ✓ Household resilience and food security
- To estimate Benefit-Cost Ratio (BCR) of the AA interventions
- To explore barriers & enablers of the AA implementation

Methods: student design and settings

- **Quasi-experimental** design to estimate AA impacts
- **Mixed** methods research:
 - ✓ Quantitative HH surveys – four-arm study design (288 HHs each arm):
 - animal health only
 - feed only
 - full-package (combination of both)
 - no intervention
 - ✓ Qualitative KIIs with beneficiaries, experts (DAs, animal health experts), community leaders, local authorities & implementing partners

Methods: indicators, data collection and analysis

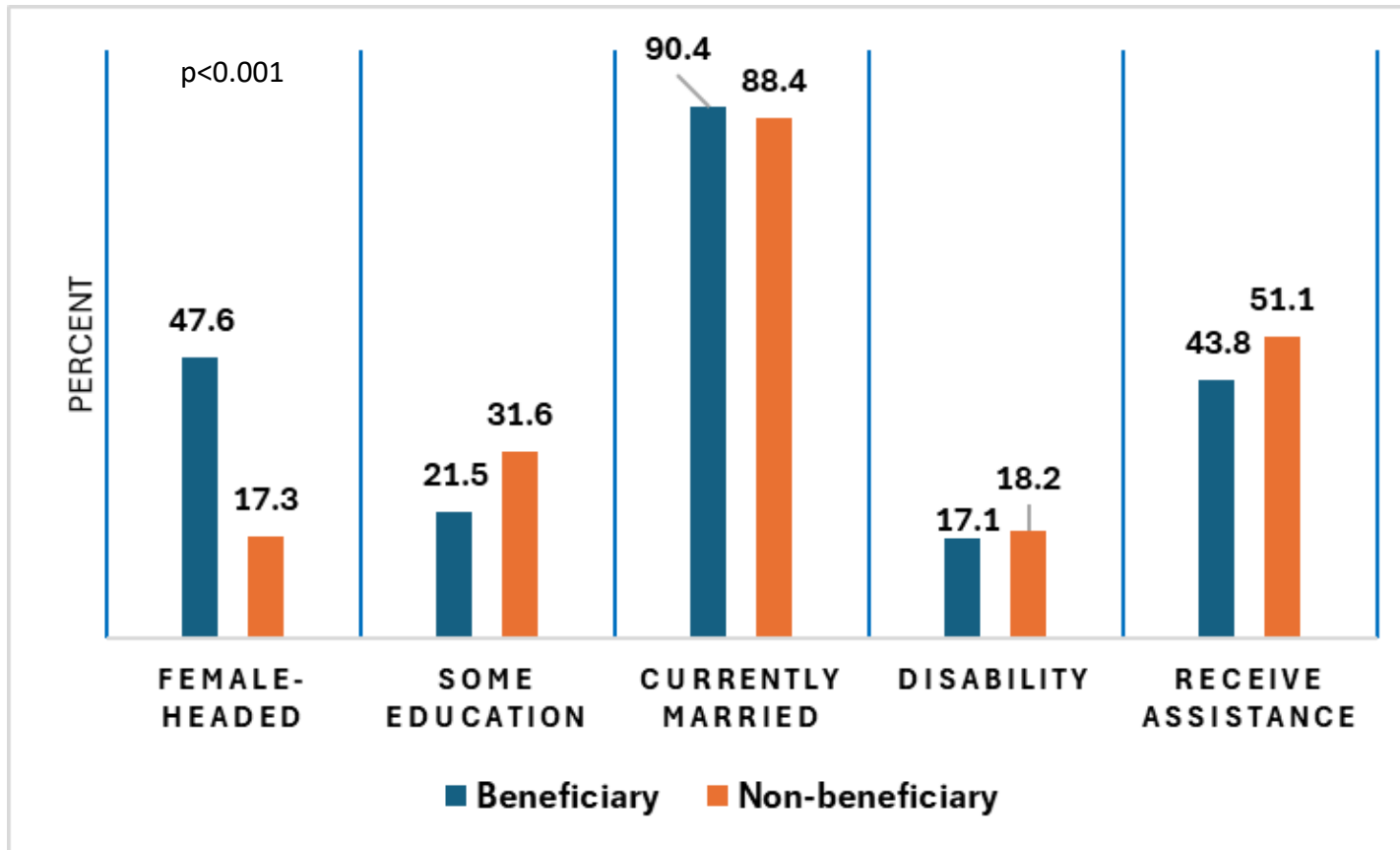
- **Outcome indicators:**

- ✓ Milk production: average volume of milk produced per day per animal
- ✓ Livestock mortality rate: total deaths/total livestock population (in 6 months)
- ✓ Food Consumption Score: food security status (poor, borderline & acceptable)
- ✓ reduced Coping Strategy Index: minimal food insecurity – stressed – crisis or worse

- **Data collection and analysis:**

- ✓ Standard and validated questionnaires, translated into Amharic, and pretested
- ✓ Qualified enumerators, trained on survey tools (Kobo), procedures, and ethics
- ✓ Comparisons tests (t-test, ANOVA, Kruskal–Wallis) & thematic analysis for the KIIs
- ✓ Benefit-Cost Ration (BCR) – to assess economic efficiency of the AA intervention

Results: baseline characteristics



- Baseline balance between beneficiary and non-beneficiary, except HH headship
- The project effectively targeted vulnerable female-headed HHs

Results: the anticipated drought prediction was accurate

- 98% HHs reported lower rainfall than previous years

"Last season, despite limited rainfall, we were at least able to plant crops. This year, however, we are left waiting and hoping for rain from Waaqaa (God), as agricultural activities have not started at all." Adama Roba (KII), a beneficiary from Oromia

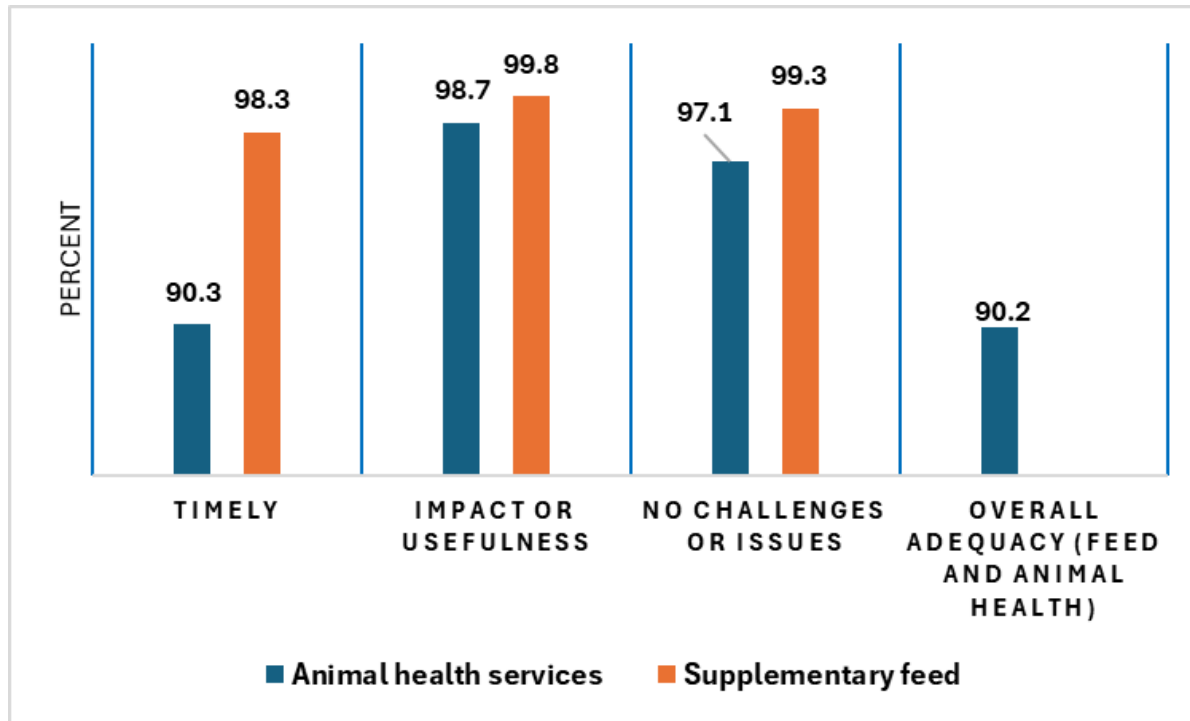
Key Informant Interview



"Rainfall was very poor; it rained only twice and ceased earlier than expected in our area, leaving no grass available for the livestock." Ibado Cafi (KII), female beneficiary from Somali

"...the last Gu season was the worst drought we have had since 2019."
Mohamed Abdimali (KII), PDB Somali

Results: interventions were timely, useful, and minimal challenge

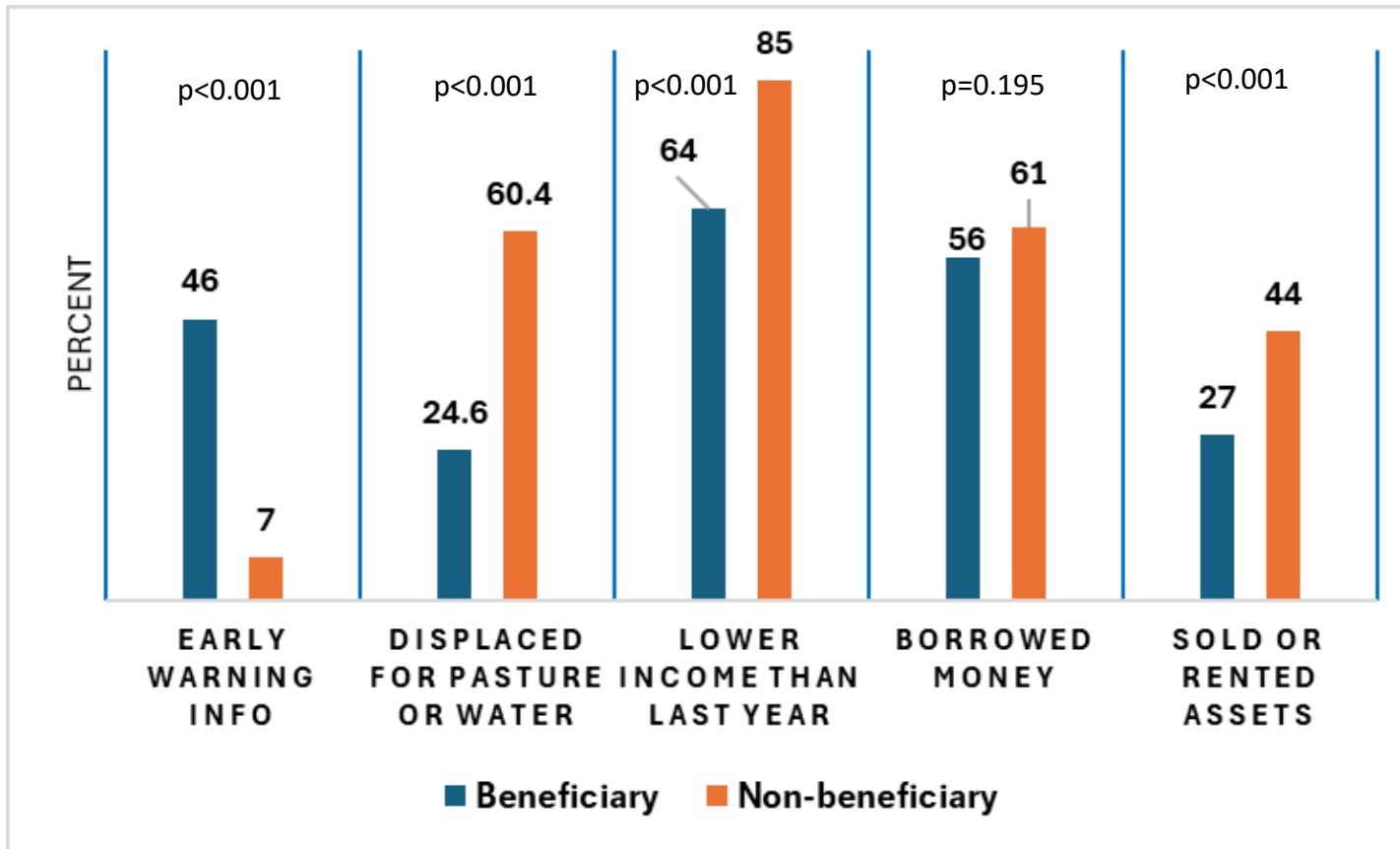


"...the previous year's saw high livestock mortality due to the absence of mass vaccination. This year, no cattle deaths were reported thanks to the timely arrival of vaccines." Adam Roba (KII), a beneficiary from Oromia

"We received the support (feed and vaccine) while our livestock were very weak. If we had not received the support, we could have lost our animals." Hussein Yusuf (KII), a beneficiary from Somali

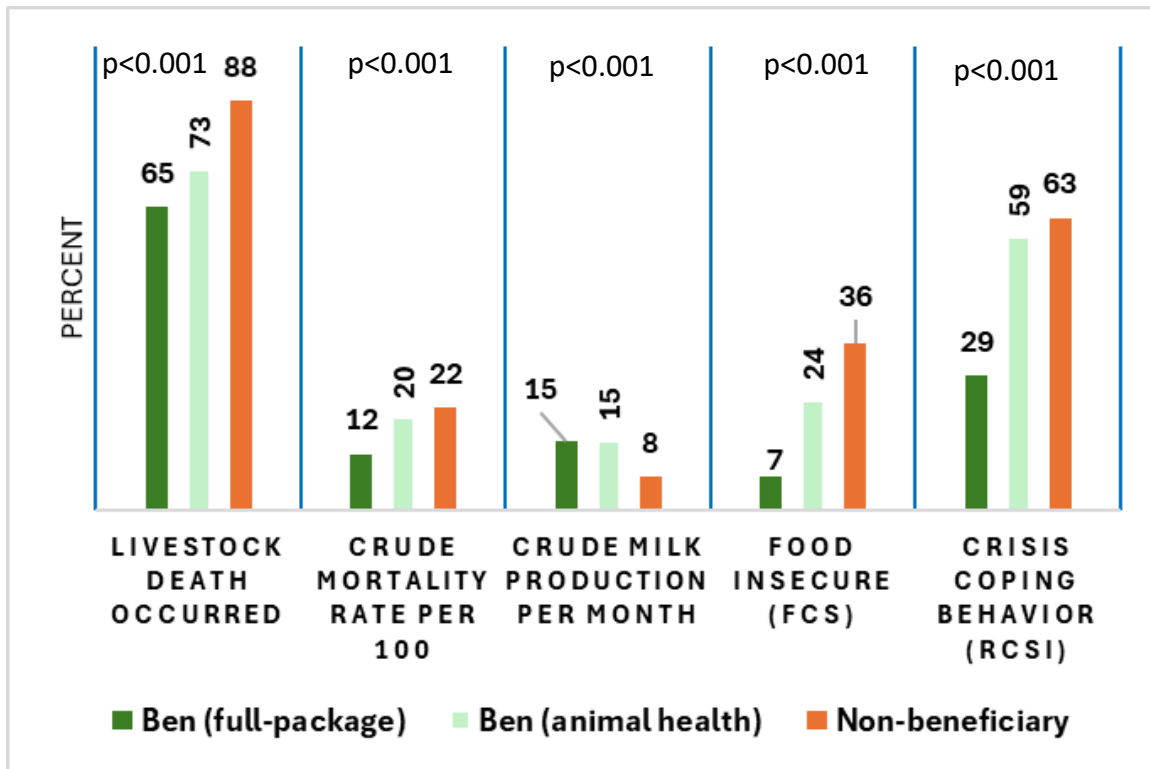
- Minor delays reported due to logistic challenges (remote areas), vaccine availability & coordination with the regional vaccine plan

Results: the AA intervention contributions



- Beneficiaries:
 - ✓ Better received early warning info
 - ✓ Less displaced in search for pasture or water
 - ✓ Less income reduction than last year
 - ✓ Less likely to borrow money
 - ✓ Less likely to sale or rent assets
- Δ TLU between 2024 & 2025 was less for beneficiaries (**-0.91 vs -2.08**; $p < 0.001$)

Results: the AA intervention contributions [2]



- Beneficiaries reported (especially full-package):

- ✓ Less livestock death
- ✓ Lower crude mortality rate
- ✓ Better milk production
- ✓ Better food security

“... Without the AA interventions, we would have seen a much higher rate of livestock loss.” Dr. Mohamed (KII), regional PDB

“Although milk production has dropped significantly, livestock body has been maintained thanks to vaccinations provided through ongoing development activities.” Obbo Kasim (KII), from VSF Oromia

Results: the AA intervention contributions [3]

- AA reinforced intervention effectiveness, strengthened community engagement and relationships, enhanced disease surveillance

*"When the livestock body condition becomes better and recovered, new livestock **fattening and trading companies emerged**." Jimale (KII), a woreda project focal from Somali*



*"...since animal health workers trained for **disease surveillance** and reporting, the monthly regional reports increased from 42% to 87.5%." Dr. Mohamed Liben (KII), regional PDB*

*"There were no tensions or conflicts among the community as the support was **timely and sufficient**. Those who missed the registration due to absence and did not receive the support were **helped by others who shared** what they had received." Abdi Ahmed (KII), a beneficiary from Somali*

Results: barriers and enablers of AA implementation

- **Barriers include:**

- ✓ Inaccessible remote areas: due to infrastructure & logistic constraints
- ✓ Seasonal migration of pastoral HHs – some has moved before support arrive

- **Enablers include:**

- ✓ Pre-campaign awareness activities helped mobilize communities
- ✓ Improved vaccine availability & coverage – enabling full-dose provision to all eligible livestock (no one was left out of the support)
- ✓ Collaboration among local experts, elders, and veterinary staff strengthened community participation and trust

Results: benefit-cost ratio

<i>Benefits (USD per HH)</i>	
Animal mortality (Adult)	157.2
Body conditions	39.5
Milk benefit	39.0
Infant animal mortality	17.4
Total benefit per HH	253.1
<i>Total costs per HH</i>	69.7
BCR (Total benefit per HH / Total costs per HH)	3.6

- \$1 investment in AA intervention generated **\$3.6 in benefits** for full-package beneficiaries:
 - ✓ Driven by avoided mortality, improved body condition, increased milk production & reduced infant livestock mortality

“We saved an expense of about Birr 20,000 (~135 \$) that we would have used to spend on livestock feed.”
Abdi Ahmed (KII), a beneficiary from Somali

Conclusions

- The AA interventions:
 - ✓ successfully targeted vulnerable HHs (female-headed HHS – 48% vs 17%)
 - ✓ enabled beneficiaries stay in their place despite drought impact (25% vs 61%)
 - ✓ helped preserve income and livestock, lowered borrowing, prevented asset sales
 - ✓ decreased livestock death (up to 45%) & increased milk production (up to 150%)
 - ✓ improved household food security (food insecurity prevalence fell up to 79%)
 - ✓ helped strengthen broader community engagement and relationships, and enhancing disease surveillance
 - ✓ benefits outweigh its cost: \$1 investment led to \$3.6 in benefits
- The intervention was timely, effective & met community needs

Recommendations and Limitations

- To enhance the efficiency of AA interventions, future programming should consider the following key areas:
 - ✓ Provision of integrated support for high impact (feed, vet services, cash etc)
 - ✓ Provision of services aligned to the needs, such as camel vaccination & treatment
 - ✓ Special consideration for hard-to-reach areas
 - ✓ Strengthen data collection and analysis to support informed decision-making
- Limitations:
 - ✓ No baseline – affects causal inference. But the comparison group is matched based on key variables and supported with qualitative evidence to show the casual link
 - ✓ Universal vaccination coverage limits identification of appropriate counterfactual



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Thank you!



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